

Executive Insight Report

Comprehensive analysis of healthcare performance and strategies

Executive Insight Report – Green Valley Medical Center



Integrated Healthcare Analytics

Combines patient outcomes, treatment effectiveness, operational efficiency, and costs into one strategic view.

Executive Decision Support

Focuses on actionable insights and business impact tailored for executive leadership decision-making.

Strategic Performance Evaluation

Enables quick assessment of strengths, concerns, and improvement opportunities within healthcare operations.

Value-Based Healthcare Transition

Supports the shift toward value-based care through data-driven improvements in outcomes and efficiency.

Project Objective



Comprehensive Healthcare Analysis

Evaluates outcomes, treatments, efficiency, and cost using Excel dashboard.

Data-Driven Decision Making

Translates raw data into actionable insights for clinical and operational improvement

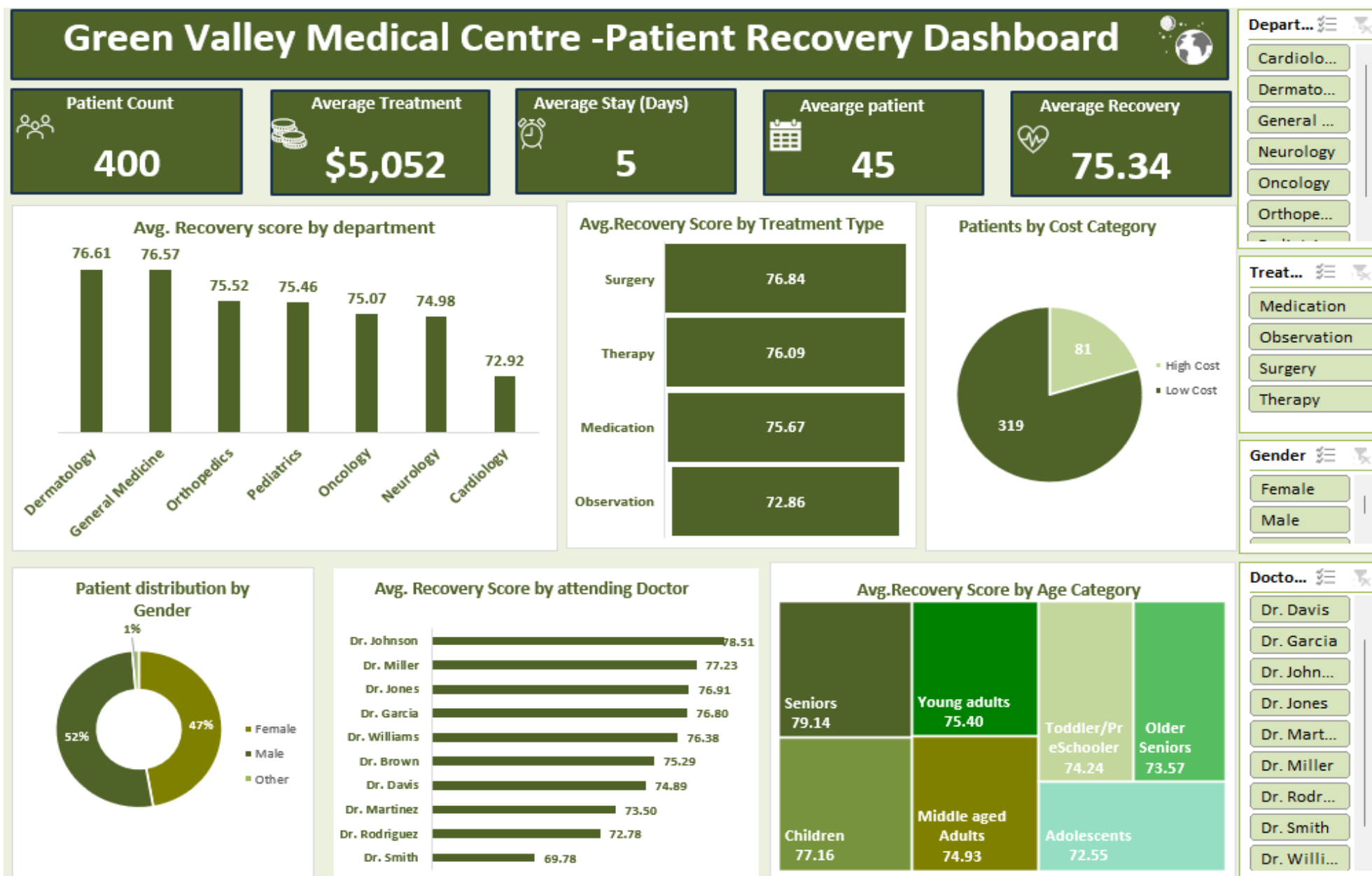
Optimizing Resources and Outcomes

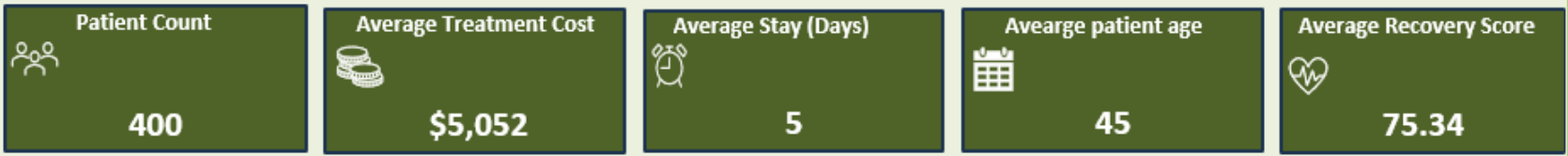
Identifies high-performing treatments and efficiency opportunities to balance care quality and cost

Business Problem



- Hospital data exists but is not effectively leveraged
- Raw data makes it challenging to generate actionable insights
- Decision-makers require data-driven insights to improve clinical outcomes and operational





Dataset Overview

Analysis includes 400 patient records providing robust data to identify meaningful trends.

Financial and Operational Metrics

Average treatment cost is \$5,052 and hospital stay lasts about five days, reflecting financial and operational performance.

Patient Recovery Score

Average recovery score is 75.34, indicating baseline clinical outcomes across the population.

Average Hospital days

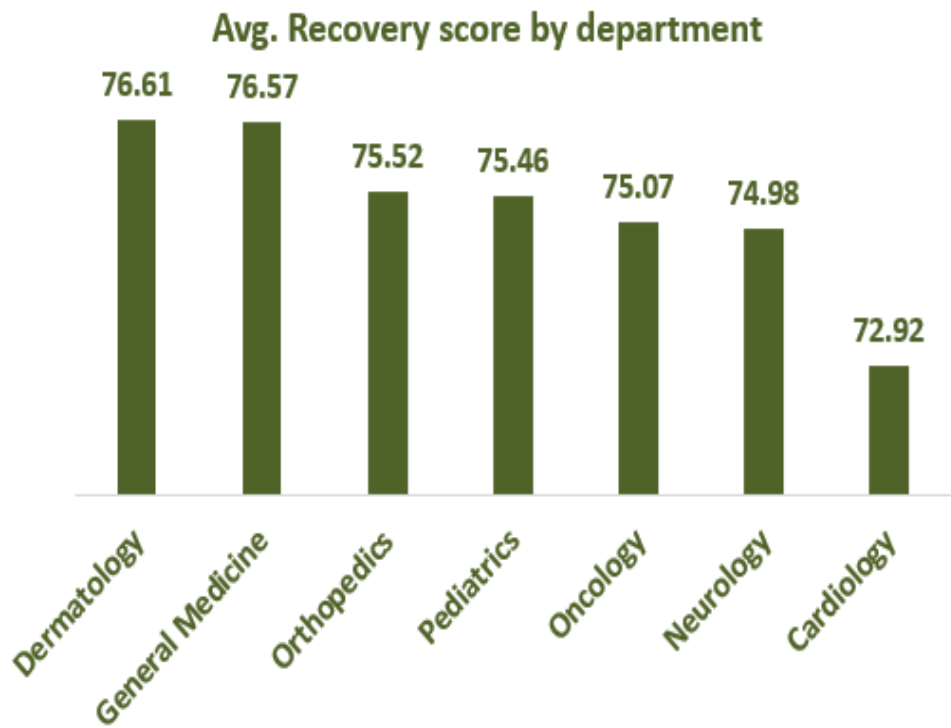
Average stay of 5 days suggests efficient patient flow with potential optimization opportunities.

Patient Age

Average patient age of 45 indicates treatment strategies should balance recovery quality and efficiency for a working-age population



Average Recovery Score by Department



Variation in Department Outcomes

Recovery outcomes vary between hospital departments due to differences in clinical practices and operational efficiency.

Recommendation:

Opportunity for Standardization

Standardizing processes using high-performing departments as benchmarks can reduce performance gaps across the organization.

Improvement Initiatives

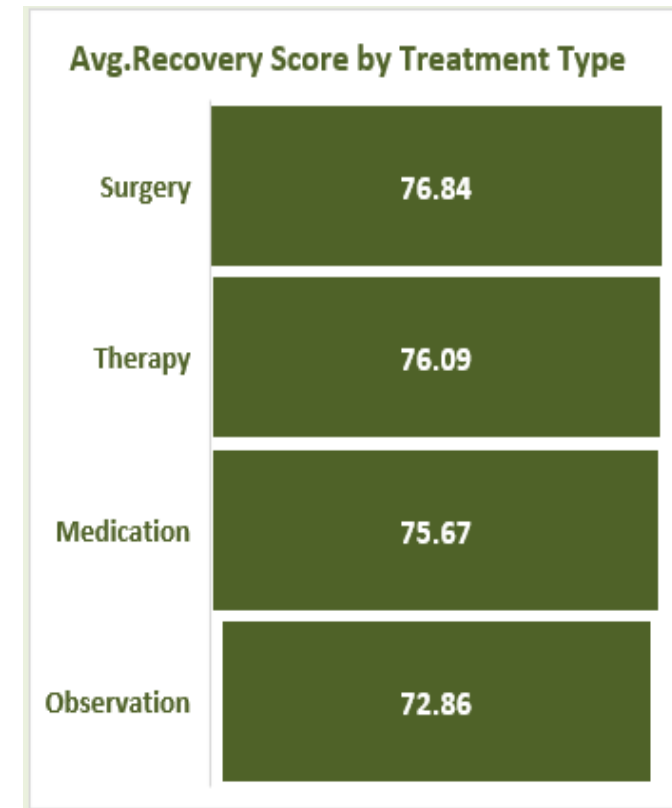
Targeted initiatives and documenting successful workflows in underperforming areas drive quality, consistency, and efficiency.

Average Recovery Score by Treatment Type

- Surgery (76.84) and Therapy (76.09) have highest recovery
- Observation has lowest recovery (72.86)
- Active treatments drive better outcomes

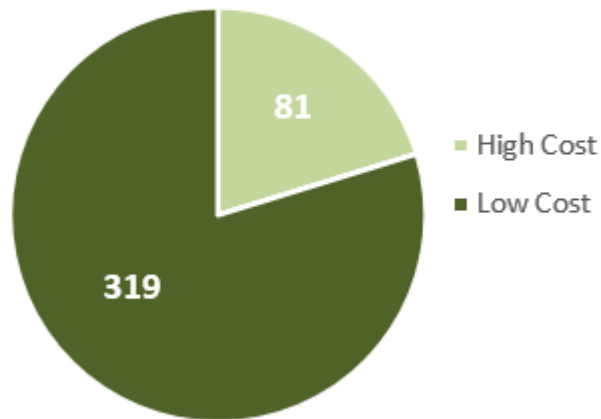
Recommendation:

Prioritize effective treatments, ensure surgical treatment rigour since it is more invasive, expensive and marginally more effective than therapy, less than 1pt difference from therapy



Average Recovery score by cost category

Patients by Cost Category



- High-cost: 77.68 vs Low-cost: 74.75
- Minimal difference in recovery outcomes
- Cost does not strongly determine success

Recommendation:

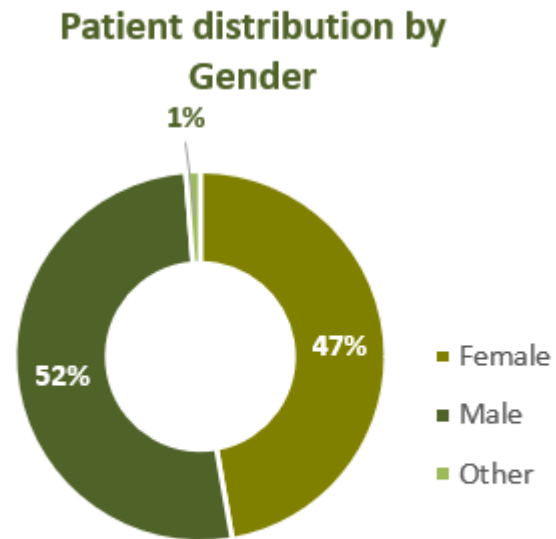
Opportunity for Cost Efficiency

Identifying lower-cost treatments with strong outcomes provides a chance to improve financial sustainability without reducing care quality.

Value-Based Care Focus

Shifting to value-based care models emphasizes maximizing patient outcomes relative to treatment costs for better strategic results

Recovery Score by Genders

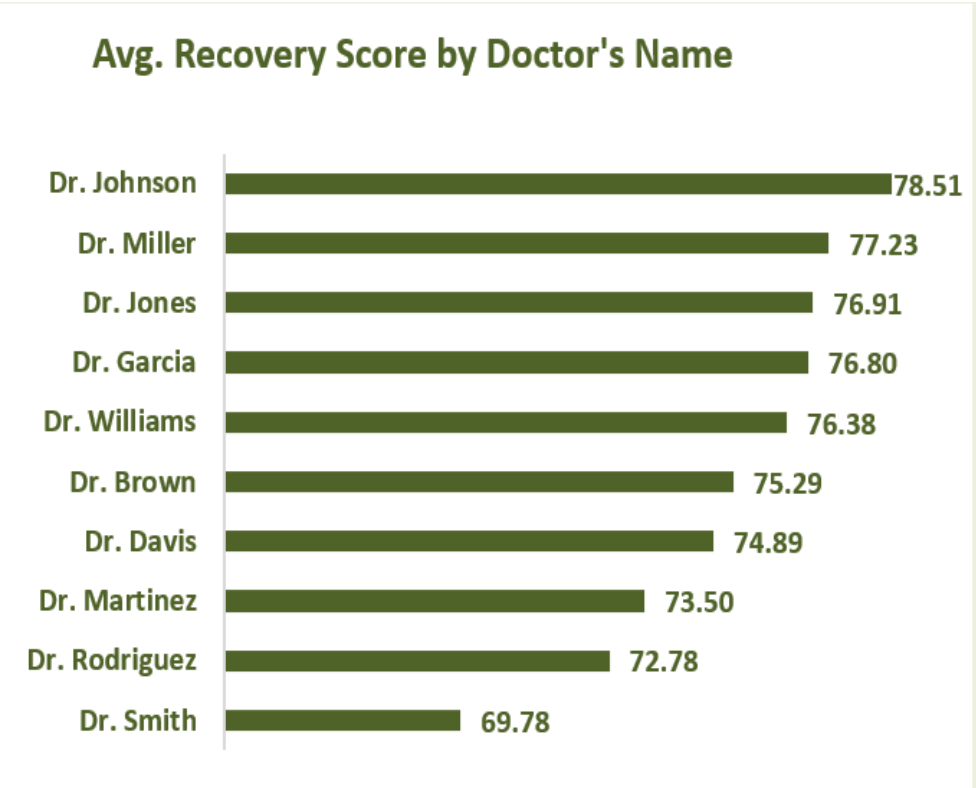


Other gender recovers 6.7 pts below average

Recommendation:

Equity audit recommended

Avg. Recovery Score by Doctor's Name



Impact of Provider Expertise

Dr Johnson had the highest average recovery Score (78.51) while Dr Smith had the least(69.78) , ~8.7 points difference.

Although physicians operate within a rotational care model, this variation persists, indicating that differences in clinical approach, decision-making, or practice patterns may influence recovery outcomes beyond patient assignment alone.

Recommendations:

Scaling Clinical Excellence

Top provider performance can be scaled through mentoring, knowledge sharing, and standardizing proven methods.

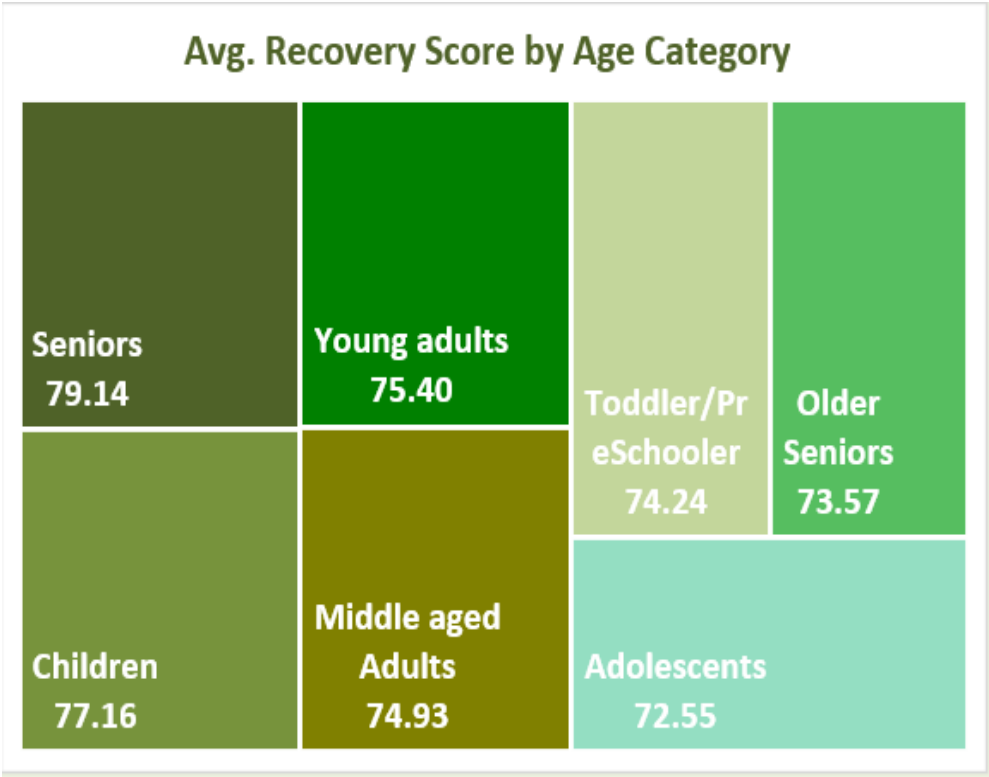
Leadership and Performance Transparency

Establishing mentorship programs and promoting transparency around performance metrics drives organization-wide improvement.

Fostering a Quality Culture

Standardized care protocols reduce variability and reinforce a focus on quality and continuous learning.

Average Recovery Score by Age Category



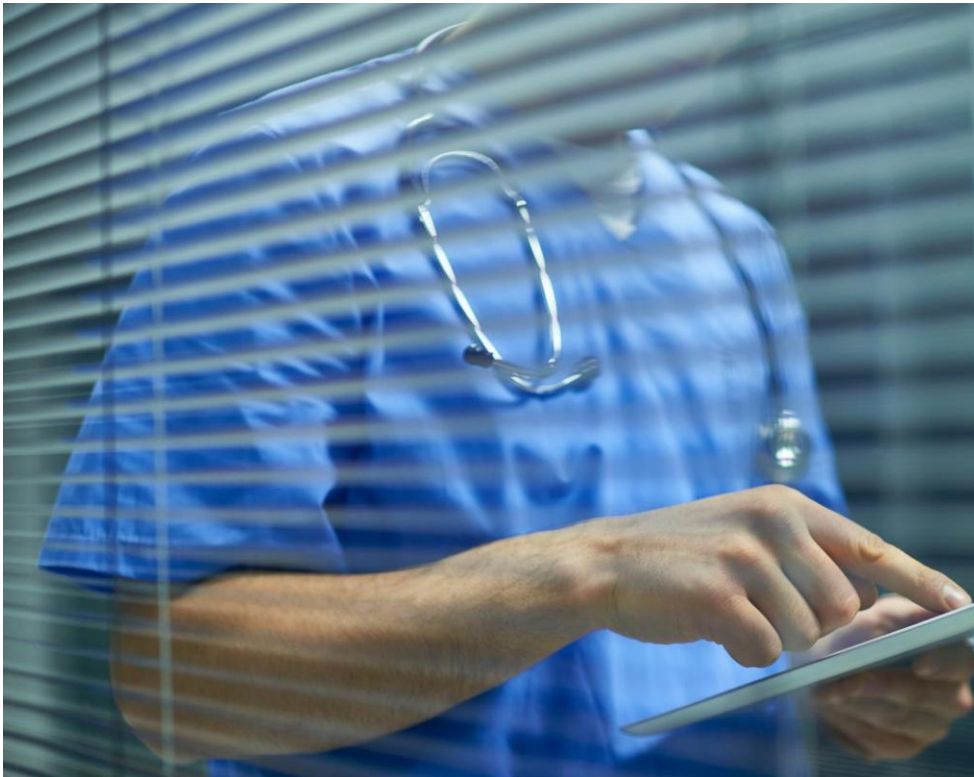
Benchmarking Length of Stay

- Seniors have the highest recovery score(79.14), this is a positive outcome given that aging generally presents typically with clinical complexity
- Adolescents presented with least average recovery score which is ~6.59 pts below the highest ranked.

Recommended:

Opportunity to evaluate and enhance age-specific clinical pathways for adolescents.

Conclusion



Improved Patient Outcomes

Analytics enable simultaneous improvement of patient outcomes, operational efficiency, and cost optimization.

Key Performance Drivers

Treatment effectiveness, best practices, provider excellence, and operational discipline drive healthcare performance.

Data-Driven Leadership

Executive insights support data-driven decisions aligning clinical quality with financial sustainability.

Continuous Improvement

Ongoing analytics use helps monitor progress, adapt strategies, and sustain healthcare improvements over time.

Thank you